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[RUNNING HEAD] M. C. T. D. Belotti et al.

[RUNNING HEAD] Pairing weather radars with eBird data for bird monitoring

RESEARCH ARTICLE

Pairing weather radars with eBird data for large-scale monitoring of swallow and martin communal roosts

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ABSTRACT

During their nonbreeding period, many species of swallows and martins (family: Hirundinidae) congregate in large communal roosts. Some of these roosts are well-known within local birdwatching communities; however, monitoring them at large spatial scales and with day-to-day temporal resolution is challenging. Community-science platforms such as the Purple Martin Conservation Association's project MartinRoost and eBird have addressed some of these challenges by centralizing data collected from regional communities. Additionally, due to the high densities of birds within these aggregations, their early morning dispersals are systematically detected by weather radars, which have also been used to collect data about roost timing and location. An important issue, however, limits spatiotemporal scope of previous radar-based studies: finding the roost signatures on millions of rendered reflectivity images is extremely time-consuming. Recent advances in computer vision, however, have allowed us to reduce this effort. The rise of this technology makes it necessary that we assess whether our biological definition of a roost matches what the machine-learning models are capturing. We do so by comparing eBird detections of roosts in the Great Lakes region with those obtained by a human-supervised machine-learning model from 2000 to 2022. With more than two decades of data, we assess the ability of these two tools to detect roosts on a day-to-day basis, and we compare the phenology of dispersals to investigate whether radar detections correspond to swallow and martin roosts or if they are associated with other well-known birds that form large aggregations. Our comparison of these datasets strongly suggests that swallows and martins are responsible for the dispersals we observe on the radars from July to late September; however, the alternative species we examined could be causing some of the detections in October.

Keywords: avian migration, communal roosting, Hirundinidae, phenology, ornithology, weather radars

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LAY SUMMARY

- Weather radars can collect the timing, location, and size of dispersals of large aggregations of birds, but identifying the species within these aggregations remains a challenge.
- We used new technology to build a dataset of 23 years of roost dispersals observed by the weather radars between July and October in the Great Lakes region. We also collected eBird checklists within the same period and region reporting >1,000 individuals of at least one of a list of species known to join large aggregations in our study region.
- Our comparison of these datasets strongly suggests that swallows and martins are responsible for the dispersals we observe on the radars. Other species known to roost in large numbers, such as *Quiscalus quiscula* (Common Grackle), *Sturnus vulgaris* (European Starling), *Turdus migratorius* (American Robin), and *Agelaius phoeniceus* (Red-winged Blackbird), could cause some of the detections in late October, but they are captured less consistently than swallows and martins.

Combinando radares meteorológicos e dados do eBird para o monitoramento em larga escala de grandes dormitórios de andorinhas

RESUMO

Durante seu período não reprodutivo, muitas espécies de andorinhas (família: Hirundinidae) reúnem-se em grandes dormitórios comuns que chegam a agregar centenas de milhares de indivíduos. Estes dormitórios são bem conhecidos dentro das comunidades regionais de observação de aves, porém permanece um desafio estudá-los em escalas espaciais mais amplas e com melhor resolução temporal. Recentemente, plataformas de ciência comunitária, como o eBird e o projeto MartinRoost da Purple Martin Conservation Association, têm abordado alguns desses desafios ao centralizar os dados coletados pela comunidade. Devido às altas densidades de aves nessas agregações, suas dispersões matinais são detectadas sistematicamente por radares meteorológicos próximos, que também têm sido usados para coletar dados a respeito de localização, data, e horário da dispersão coletiva dos dormitórios. Até recentemente, no entanto, encontrar os dormitórios em milhares de imagens obtidas diariamente pelos radares permanecia uma limitação dos estudos que usam esta metodologia. Avanços recentes em visão computacional e computação de alto desempenho, contudo, nos permitiram reduzir significativamente este esforço. O surgimento dessa nova tecnologia torna necessário avaliar se nossas definições conceituais e biológicas de um dormitório correspondem ao que é capturado pelos modelos de aprendizado de máquina. Para isso, comparamos as detecções de dormitórios no eBird na região dos Grandes Lagos com aquelas obtidas utilizando um modelo bem estabelecido de visão computacional capaz de encontrar automaticamente as dispersões em dados de radar. Com mais de duas décadas de dados, avaliamos a capacidade destas duas ferramentas de detectar dormitórios e comparamos a fenologia das detecções com o fim de investigar se os radares estão capturando andorinhas ou outras espécies de aves que também formam grandes agregações. Nossos resultados sugerem que os dados dos radares oferecem uma

cobertura espacial e temporal mais ampla do que a oferecida pelo eBird. No entanto, principalmente em outubro na região dos Grandes Lagos, quando outras espécies também se reúnem em grandes agregações, a falta de resolução taxonômica dos radares pode limitar o uso destes dados. Novas metodologias que integram dados de radar com observações de ciência comunitária podem ajudar a mitigar alguns destes problemas, particularmente em estudos de fenologia.

Keywords: migração de aves, dormitórios comunais, Hirundinidae, fenologia, ornitologia, radares meteorológicos

INTRODUCTION

Swallows and martins (family: Hirundinidae) are highly specialized aerial insectivores which are mostly migratory, with some species traveling up to 20,000 km between their breeding and nonbreeding grounds. During the nonbreeding period, many species of this family congregate in large communal roosts that can gather hundreds of thousands of individuals (Ward and Zahavi 1973, Winkler 2006). These large aggregations can offer protection against predators and may act as information-sharing centers, where birds gather to learn about optimal foraging locations. The latter theory about communal roosting behavior is particularly well-suited for swallows and martins, because these central-place foragers rely on ephemeral but locally abundant emergences of aerial insects (Lalla et al. 2022, Ward and Zahavi 1973, Winkler 2006). In the Americas, the timing and location of these aggregations are often predictable, with roosts occurring on the same site year after year. Roosts gathering *Riparia riparia* (Bank Swallow), *Tachycineta bicolor* (Tree Swallow), and *Hirundo rustica* (Barn Swallow), for example, have been reported as early as 1918 in Black River Bay, on Lake Ontario, USA, where our data suggests they are still observed to this day (Bent 1942; Belotti et al. 2023).

Monitoring these aggregations is crucial for the conservation of these species which, like other aerial insectivores, have been facing widespread breeding population declines for reasons not yet completely understood (Rosenberg et al. 2019, Spiller and Dettmers 2019). Furthermore, understanding the drivers and constraints that dictate roost size and timing could help us elucidate questions related to the ecological function of these roosts and their downstream effects on regional food webs. However, in spite of the size of these aggregations, finding them can be challenging. Birds usually arrive at the roosting location in relatively low concentrations at dusk, when lighting conditions are limited. Additionally, their much denser dispersal occurs early at dawn, before most human observers typically start their daily activities (Russell and Gauthreaux 1999, Arruda 2023). In densely populated areas, roost observations have grown in the past century, particularly with the rise of community-science initiatives. In remote areas of the Amazon Rainforest, however, it is likely that some of these large gatherings of swallows and martins remain all but unnoticed by human eyes.

The Purple Martin Conservation Association (PMCA) hosts the most comprehensive dataset of roost locations in North America. Through their project MartinRoost, they collect reports of *Progne subis* (Purple Martin) roosts submitted by volunteers throughout the United States. Each report corresponds to roost observations from one season, with start and end date, location, species composition, notes about observations made on the ground, and science dissemination opportunities associated with that roost—some roosts have become regional events, with roost watch parties gathering hundreds of families. In the past 20 yr, the community-science platform eBird has grown as yet another roost monitoring tool in North America

(Sullivan et al. 2009). Birdwatchers on this platform submit checklists of observed birds at a given location, at a specific time and date. They also have the option of reporting the number of individuals observed for each species. Even though community-science data can be used to obtain valuable ecological insights, analyzing these data poses some important challenges. Because the sampling process is heterogeneous, there is significant variation in observation time, day of the year, and effort. Additionally, since remote regions can be difficult to access, community-science datasets usually have a strong spatial bias towards regions with denser human populations (see Johnston et al. 2021 for a review).

Weather radars can address some of the sampling biases that permeate community-science datasets. Due to the high densities observed in roosts and the characteristic dispersal pattern of swallows and martins, their early morning departures produce a distinctive doughnut-shaped signature on data collected by nearby weather radars, which can be used to quantify aggregation size, location, date, and timing (Russell et al. 1998; Russell and Gauthreaux 1998, 1999). Previous efforts to find these signatures and collect data from them have allowed Russell et al. (1998) to map 33 roost locations throughout the southeastern United States for the first time (Russell et al. 1998). We have also confirmed, by pairing radar-detected roosts with community-science reports, that swallows and martins are the main species found in these aggregations (Kelly and Pletschet 2017, Kelly et al. 2012). Additionally, roosts in the United States occur on 4 main land cover types, namely urban, forest, water bodies, and croplands, and that urban roosts are typically more persistent throughout the years when compared to roosts observed in other land cover types (Bridge et al. 2015). Lastly, Laughlin et al. (2016) quantified autumn occupancy dynamics in roosts in southeastern Louisiana and central peninsular Florida, confirming that the latter acts both as a stopover and a nonbreeding region for *T. bicolor*, whereas central Florida serves more exclusively as a nonbreeding location for this species.

Until recently, however, our ability to use radar data to obtain fine-scale spatio-temporal data on roosting behavior was limited by the fact that finding roost signatures on millions of rendered reflectivity scans is extremely time-consuming. Nevertheless, recent advances in computer vision and high-performance computing have allowed us to streamline some of this effort. A machine-learning pipeline developed by Cheng et al. (2020), capable of detecting roosts on radar data and tracking them across successive scans, allows us to leverage the broad spatial and temporal coverage of the network of weather radars to understand the communal roosting behavior. By employing this technique to find roosts, human supervision is limited to removing false positives and labeling tracks according to how much contamination from precipitation and other sources of radar noise is caught within them. This model has since been improved, further reducing the human supervision effort by improving the model's accuracy (Perez et al. 2022), and new techniques have been developed to estimate cumulative curves of roosting activity throughout the season with less manual screening (Perez et al. 2024).

These new advances allow us to gain spatial and temporal reach in our study of swallow and martin communal roosting behavior. However, they also require that we match our conceptual understanding of this behavior with what the technology we use to study it is observing and measuring. This is particularly true with regards to species identification. In our labeling protocol, we defined a series of morphological characteristics of roosts collected from the literature (Kelly and Pletschet 2017, Kelly et al. 2012, Russell et al. 1998, Russell and Gauthreaux 1999, 1998; Laughlin et al. 2013). We aim to understand if roosts selected with this protocol correspond to observations made on the ground by community scientists. We do so by comparing the timing of radar-detected roosts with that observed in eBird data for nine species

known to form large aggregations. These nine species were divided into 2 groups: the swallow and martin group, which contains *Petrochelidon pyrrhonota* (Cliff Swallow), *Progne subis*, *T. bicolor*, *H. rustica*, *R. riparia* and is the group we hypothesize to be causing most of the activity in the Great Lakes. The second group, which we will refer to hereafter as the group of “alternative species” contains *Quiscalus quiscula* (Common Grackle), *Agelaius phoeniceus* (Red-winged Blackbird), *Turdus migratorius* (American Robin), and *Sturnus vulgaris* (European Starling), which are also known to form large aggregations (Eastwood et al. 1962, Clark et al. 2020, Broeke 2022). Additionally, we assessed the quality of the spatial coverage of these two sources, which we measured by the number of roosts they can detect per day.

MATERIALS AND METHODS

Radar Data Processing

In the Great Lakes region, swallow and martin roosts typically occur in the summer, between July and August (Russell 1996, Kelly and Pletschet 2017). We downloaded and rendered all radar scans available between June 1 and October 31 for the 12 radar stations within our study region from 2000 to 2022. We focused on the period between 30 min before local sunrise and 90 min afterwards, when we typically observe roost departures. These data are publicly available in an Amazon Web Services server (<https://registry.opendata.aws/noaa-nexrad>). Radial velocity and reflectivity in these scans were used as input to the machine-learning pipeline developed by Cheng et al. (2020), which leverages the capabilities of a Faster R-CNN model coupled with a Kalman Filter to detect and track roost dispersals in radar data (Cheng et al. 2020). The pipeline outputs collections of bounding boxes (so-called tracks), each collection representing a single dispersal event. We designed a manual screening protocol (see Supplementary Material) used by 3 human annotators (MB, YD, and VS) to label each track according to the intensity and type of noise contamination within each track, if there was any. The screeners also labeled days when precipitation, anomalous propagation, or other sources of radar noise were so intense that we would not have been able to see the roosts if they were present. Because we were interested in the presence/absence of roosts for this study, we did not remove tracks with noise-contaminated dispersals.

Some of the weather radars selected for this study have overlapping ranges and it often occurred that the same roost was detected by two radars (Figure 1 for a map of the study region with the radar stations and their ranges). We account for this by grouping overlapping roosts occurring on the same day. We did this by first obtaining the union of all the bounding boxes for each dispersal. We then calculated the ratio between the area of the intersection and the area of the union of each pair of overlapping tracks occurring on the same day. If this ratio was higher than 0.2, the tracks were considered to represent the same roost dispersal. We refer to these grouped tracks as unique detections or unique radar tracks. We determined the location of each roost dispersal by obtaining the center of the first bounding box on each unique detection.

eBird Data Processing

We downloaded all the eBird checklists made from 2000 to 2022 that reported at least 1 of the 5 swallow and martin species known to join roosts in the United States, namely: *Tachycineta bicolor*, *H. rustica*, *R. riparia*, *Petrochelidon pyrrhonota*, and *Progne subis*. Additionally, we obtained eBird checklists for *Q. quiscula*, *Turdus migratorius*, *S. vulgaris*, and *A. phoeniceus* because these 5 species also form large aggregations and are sometimes reported to join swallow

and martin roosts (see Figures 2 and 3 for examples of dispersals captured by weather radars). Our initial filtering process removed checklists with the same unique sample identifier, indicating that they were made by groups of observers. We selected checklists made after 16:00, when roosts start to form, and before 10:00, when roosts disperse. These were further filtered to keep only checklists reporting >1,000 individuals, indicating large aggregations. Lastly, we removed checklists that were outside the range of 150 km of the 12 radar stations in the Great Lakes region.

If many individuals observed the same roost and were not part of a single observation group (reported on eBird with a unique sample identifier), then that single roost was associated with multiple checklists. Because we were interested in unique roost detections, we grouped checklists made at the same time within 1 km of each other. We also grouped observations made in the morning or in the evening of the previous day in the same location (within 1 km of each other), to reflect the fact that the same roost can be detected on two different days. We refer to each observation resulting from this grouping procedure as a unique mass, high-count detection, or a unique eBird roost observation.

Comparison

In our comparison of the timing of radar observations and eBird checklists, we extracted the number of unique detections per day of the year from each source and for each species in the case of eBird data. With this, we could extract the peak roost detection dates for both sources. Additionally, we calculated the cumulative number of unique detections per day for each source between June 1 and October 31, grouping checklists for all species of swallows and martins and for our group of five alternative species. We filtered eBird data from years after 2010 in order to ensure we had enough days with detections for each year. We standardized the cumulative curves dividing them by the summed count per year. We fit a logistic smoother to these data to find a single representation of the phenology of radar, swallow and martin, and alternative species detections. Finally, we performed two Pearson correlation tests: one between the smoothed curve of predictions for swallows and martins and the smoothed predictions for radar detections, and one between the smoothed curve for our alternative species and the curve for radar detections. Because these smoothed curves were temporally autocorrelated, we differenced each curve before performing the 2 correlation tests. Differencing of a time series is performed by subtracting measurements at time $t - 1$ from measurements at time t for all possible values of t . This procedure removes some of the temporal autocorrelation from a time series and ensures that it satisfies the assumptions of the Pearson correlation test.

When assessing the quality of the spatio-temporal coverage of both sources, we compared the distribution of the number of unique detections per day for each source within the same region, during the same periods. We did this by further filtering the eBird dataset to remove observations made before June 1 or after October 31, when the radar data were not processed. We also removed checklists submitted when the radars were not sampling due to heavy precipitation, anomalous propagation, other sources of unidentified noise, or files missing from the archive. Finally, when roosts are located within 5 km of the radar station, they do not appear with the characteristic ring shape, because the radar samples them at very low elevations. This causes the machine-learning pipeline to fail to detect them. Because of this, we omitted unique eBird detections made within 5 km of the radar station.

RESULTS

After our post-processing of eBird data, we were left with 788 unique detections of swallow and martin aggregations made throughout the year, from 2000 to 2022, regardless of radar sampling. Of those, 94% were associated with checklists reporting high counts of a single species: 256 (32% of total) reported roosts gathering only *Progne subis*, 211 (27%) detections were of *Tachycineta bicolor* aggregations, 164 (21%) were *R. riparia* roosts, 86 (11%) were roosts containing exclusively *H. rustica*, and 19 (2%) unique detections reported aggregations of *Petrochelidon pyrrhonota* alone. The remaining 52 (7%) unique detections referred to checklists reporting high counts of multiple swallow and martin species. Most of the swallow and martin checklists with high counts were made during our sampling window from June 1 to October 31 when the radars were sampling, that is, on days where the data were available and not fully contaminated by weather or anomalous propagation ($n = 583$, 74%, Figure 4).

The seasonality of the number of detections per day for both sources was similar for these species; however, eBird detected some large numbers of swallows and martins between April and June, before the start of the roosting season in July (Figure 4). Large counts of *Q. quiscula*, *S. vulgaris*, *T. migratorius*, and *A. phoeniceus* have a very different seasonal pattern from that observed for swallows and martins, with two peaks of detections: one before and one after their breeding season (Figure 5). The analysis of the cumulative detection curves for radar data and for our two groups of study species shows how the phenology of radar detections matches what we observe for swallow and martin high count checklists (Pearson correlation coefficient 0.79, p -value < 0.001 , $n = 152$) and is significantly different from the phenology of the group of alternative species (Pearson correlation coefficient -0.24 , p -value = 0.001, $n = 152$; Figure 6).

We found that the average number of unique detections per day for the radars was 8.31 (SD = 7.10), while the average number of eBird unique detections of swallows and martins per day was 1.55 (SD = 1.00, Figure 7). The maximum number of unique detections made by the radars on a single day ($n = 38$) was 4.75 times larger than the same metric for eBird ($n = 8$). The radars detected more roosts than eBird on 95% of the sampled days (Figure 7). Each year, the maximum number of unique eBird detections reported on days sampled by the radars (the peak detection date per year) was reached between July 16 and August 29 across all years with more than 2 unique detections per day ($n = 10$). The median peak detection date occurred on August 16 (Figure 4). The peak of detections across all years ($n = 23$) for the radars varied between August 4 and August 24, with an average number of detections at the peak dates of 27.4 (SD = 4.2). The median peak date across all years occurred on August 15 and 17 (Figure 4).

With regards to the last report of the roosting season, we found some variation across species on eBird data. The last report of the season for *R. riparia* occurred between July 29 and August 28, with a median between August 20 and 21. For *H. rustica*, the last high-number report occurred between August 4 and September 16, with the median on August 26. The last report of *Progne subis* roosts occurred between August 27 and September 15 across all years, with the median in early September, between September 6 and 8. Lastly, for *Tachycineta bicolor*, the last report occurred between September 6 and October 26, with the median between October 10 and 12 (Figure 4).

DISCUSSION

We summarized the phenology of eBird high-count detections across 23 years into a single curve for each of 9 species of birds known to form large aggregations in the Great Lakes region of North America. We compared the temporal pattern of eBird mass detections with what we obtained from radar detections to assess which species could be responsible for the patterns

observed on local radars from July to October. The phenological similarities between the pattern of radar detections and swallow and martin eBird detections across years suggest that these species are likely the ones joining the radar-detected aggregations, indicating that the machine-learning model, paired with human supervision, is actually capturing the communal roosting behavior observed in previous studies (Kelly and Pletschet 2017, Kelly et al. 2012). However, some of the activity observed in the later part of the season is likely driven by *A. phoeniceus* and *Q. quiscula*, which would also explain why the radar detections per day never return to the zero baseline after the roosting season peaks. Further studies need to take this into account, particularly when examining phenology changes at the end of the roosting season.

The literature would benefit from a comparative study of roost dispersals of all species known to join these aggregations. Different species may disperse at different times of the day or produce varying signatures on the radar reflectivity images, as is suggested by Figure 2. Additionally, examining other radar products, such as radial velocity and differential reflectivity, could provide insights into whether these measurements can capture species-specific differences in flight speed and aspect ratio, respectively (for a review of these products, see Stepanian et al. 2016). Ultimately, automatic roost identification systems like the one proposed by Perez et al. (2022), which was used to build our Great Lakes dataset, may not only tell us where roosts are, but also which species are aggregating within them.

Interestingly, we observed large aggregations of all species but *Progne subis* before the main post-breeding roosting pulse. The nature and function of these aggregations is not yet well-established. These are unlikely to be breeding colonies, both because of the large counts and because of the brief pause in occurrence before the post-breeding roosts start. On the other end of the season, we notice that *T. bicolor* remain roosting until late October, unlike all other species of swallows and martins. In fact, eBird data suggests that *R. riparia* are the first to disappear from the roosts at the end of August, followed by *H. rustica*, who also stopped being detected in late August. *Progne subis* are the next to leave, in mid-September, and finally *T. bicolor* typically stay for another month, and only stop being detected in mid-October. On eBird data, the end of the roosting season for swallows and martins is not confounded with the start of the aggregations of the group of alternative species. For that reason, we suggest that the two data sources could be integrated in phenology models to obtain a more accurate definition of when communal roosts of swallows and martins stop occurring every year, much like what was done in Horton et al. (2018) and in Horton et al. (2019).

Establishing the true peak roosting date of swallows and martins from eBird data alone might be challenging. In most years, the daily peak number of unique detections did not exceed 3, which leaves a narrow margin for over- or underestimation of the total number of roosts due to sampling biases, such as spatiotemporal changes in observer effort. This was not the case with the radar dataset, in which the average number of roosts detected at the peak date was 27.4 with a relatively low standard deviation of 4.2. Probably because of this, we also observed more variation on peak detection date throughout the years on eBird—47 days from July 16 until September 1—when compared to the range of peak detection dates on radars—20 days between August 4 and August 24. Additionally, the low standard deviation in the number of detections on peak dates seen in the radar data suggests that they are indeed detecting most roosts within their range.

Lastly, the difference between the number of roosts detected per day by eBird and the weather radars suggests that the latter can offer a much broader coverage of the airspace, helping us locate rare aggregations that are hard to find if observers are not directly looking for them.

Additionally, with the methodology established by Chilson et al. (2012), weather radars allow us to estimate the size of each communal roost systematically without running into some of the biases associated with visual counts of high numbers of birds (Davis et al. 2022). Pairing data collected by weather radars with taxon-specific observations made on the ground can enhance our ability to monitor and study the communal roosting behavior of a wide range of species at transcontinental scales. These capabilities can be particularly important in remote, but severely threatened regions of the world, such as the Amazon Rainforest.

Supplementary material

Supplementary material is available at *Ornithological Applications* online.

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Ethics statement

This study did not involve the capture or handling of animals. Our methods do not change avian behavior. No ethical concerns to report.

Conflict of interest statement

The authors have no conflict of interest to report.

Author contributions

Maria Carolina T. D. Belotti: Conceptualization, Investigation, Methodology, Software, Data Curation, Writing Wenlong Zhao: Methodology, Software, Data Curation Victoria F. Simons: Investigation Yuting Deng: Methodology, Investigation Gustavo Perez: Methodology, Investigation Elske Tielens: Conceptualization, Methodology, Supervision Subhansu Maji: Conceptualization, Methodology, Supervision Daniel Sheldon: Conceptualization, Methodology, Supervision Jeffrey F. Kelly: Conceptualization, Methodology, Supervision Kyle G. Horton: Conceptualization, Methodology, Supervision, Writing

Data availability

The data and scripts that support the findings of this study is publicly available in Zenodo (<https://doi.org/10.5281/zenodo.15809709>). The dataset of swallow and martin roost detections is available on Figshare (<https://doi.org/10.6084/m9.figshare.20137961.v2>).

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Figure 1. Map of our study region within the Great Lakes of North America. Circles around each radar station represent areas examined for roosts. The diameter of the squares is 300 km in length. Each station is associated with a unique 4-letter code represented in the figure.

Figure 2. (A) An example roost dispersal from the dataset of the Great Lakes region of North America. (B) A dispersal of *Turdus migratorius* (American Robin) documented by Van Den Broeke (2018).

Figure 3. (A) a dispersal of *Agelaius phoeniceus* (Red-winged Blackbird) examined in Clark et al. (2020), and (B) a dispersal of *Sturnus vulgaris* (European Starling) reported on eBird checklist S11806632 (Hertz 2012).

Figure 4. Plots representing the number of unique detections of roosts per day of the year made by eBird (A) and by the radars (B) from 2000 to 2022. Plot (A) aggregates the eBird mass detections per species represented on the plots at the bottom (C–H). The plot with "Mixed Species" (H) shows numbers of unique high-count detections per day reporting at least 2 of our target swallow and martin species. Shaded areas represent the period from June 1 to October 31, when we examined data from the radars (grey), the range of peak detection dates across all years on eBird (brown), and the range of peak detection dates across all years on radars (in yellow).

Figure 5. Plots representing the aggregated number of unique mass detections of *Quiscalus quiscula*, *Agelaius phoeniceus*, *Sturnus vulgaris*, and *Turdus migratorius* per day of the year, across all years, from 2000 to 2022 (A). The two plots on the bottom (B–F) represent the same variable for each species and for unique mass detections reporting both species (mixed species). Shaded grey area corresponds to the period sampled by the radars (from June 1 to October 31)

Figure 6. Plots (A), (B), and (C) represent standardized cumulative mass detections per day of the year across all years (gray background points) and the smooth curve obtained from fitting a logistic regression to the points. Plot (D) contains just the smooth curves for comparison. The Pearson correlation coefficient between the detrended curves obtained for swallow and martin eBird detections and the radar detections was 0.79 (p -value < 0.001 , $n = 152$), whereas the correlation coefficient between radar detections and grackle and blackbird eBird high-count detections was -0.24 (p -value < 0.001 , $n = 152$).

Figure 7. Plots (A) and (B) are histograms of the number of unique detections per day for eBird (A) and for the radars (B). On (C) we plot the histogram of the difference between the number of unique tracks per day and number of unique high-count checklists per day. The number of radar detections per day was larger than the number of eBird detections per day on 95% of days.













